Byegerm2.0 Reproduction Guide

Rotating multi-tube instrument: fabrication, assembly and system commissioning

1 Technical field

Byegerm2.0 combines temperature control for multiple tube positions, rotary switching, fluorescence imaging and communication with a mobile device. This manual explains how the structure, circuit boards and control program work together, and describes how to replicate the instrument through fabrication, assembly, wiring and integrated testing.

2 Background

To address the limited throughput of the dual-tube instrument, byegerm2.0 increases capacity for settings such as kindergartens and nursing homes. The main chamber has a rotating structure with three faces, each holding five tubes. Each face rotates into the same imaging field of view in turn. A separate temperature-controlled zone at the front holds two tubes.

3 System design

The instrument measures approximately 15 × 30 × 20 cm and has six modules: main control, AI, temperature control, optics, power management and mechanical control. The RK3588 handles imaging, analysis and communication, while the STM32F407ZGT6 controls the peripherals.

The functional circuits are separate units connected by wiring harnesses with terminal connectors. This simplifies hardware assembly, troubleshooting and maintenance. The circuits are grouped by three functions, main control, peripherals and power, and implemented on the following four PCBs.

Board	Main functions
Core board	STM32 minimum system, inter-board connections, debugging and communication
Control board	Receives control signals and connects to the heating, lighting, temperature-sensing and rotary actuation components
Mechanical board	Connects the main chamber's six temperature-sensing channels, three lighting groups and heating elements
Power board	Voltage conversion, filtering and power distribution to the functional boards

4 Figures

Figure 1 shows an exploded view of the modules. Figures 2 to 4 show the core circuits and main control boards. Figure 5 shows the actuator drive circuits. Figures 6 and 7 show the mechanical board, and Figures 8 and 9 show the power board. Figure 10 covers the temperature-sensing and power interfaces. Figures 11 and 12 show photographs of the supporting modules. Figure 13 shows the workflow between the app and instrument, and Figure 14 shows photographs of integrated system testing.

Prepare the structural parts, four PCBs, computing board, temperature-control and optical components, actuators, battery and management components. Also prepare screwdrivers, a multimeter, soldering tools and tools for preparing the wiring harnesses.

5 Implementation

5.1 Structural fabrication and component installation

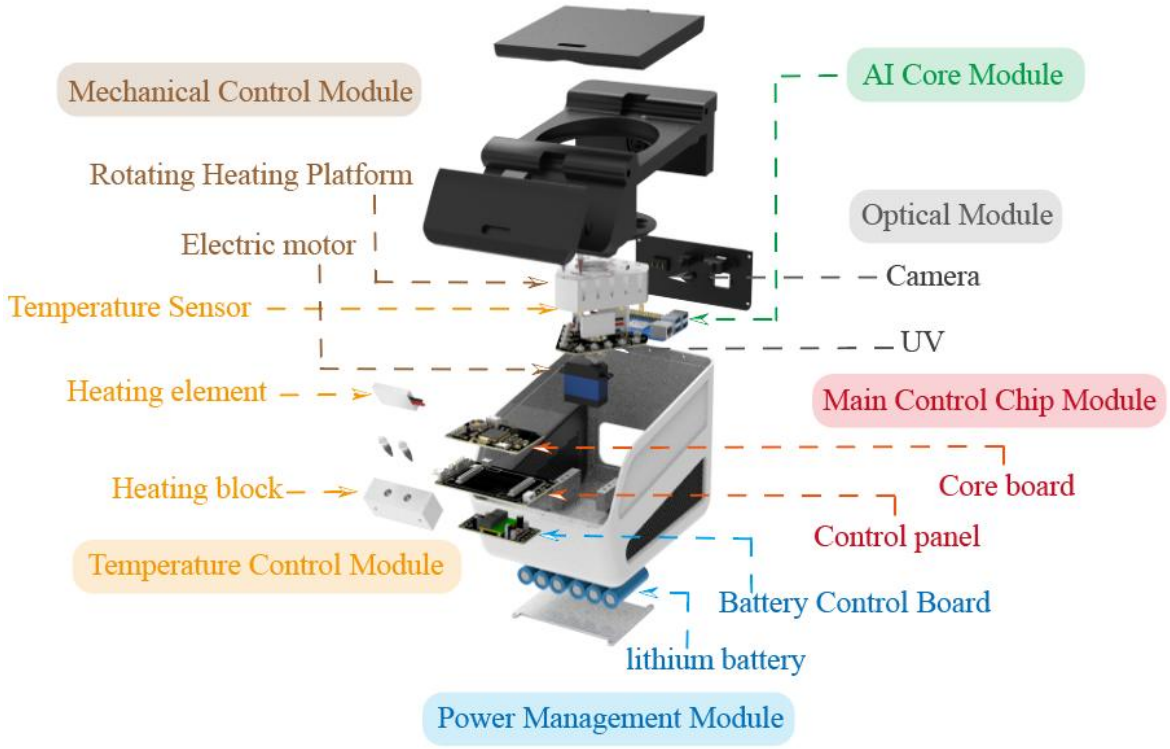


Figure 1 Assembly arrangement of the front temperature-controlled zone, rotating working chamber and internal functional modules

1 Fabricate the structural parts. Create the models in SolidWorks 2024 and check for assembly interference. Import the models into PrusaSlicer to generate the print files. Print the structural parts in PETG on an SK1 FDM printer, remove the supports and perform a trial assembly. Install insulation in the heating areas as designed to prevent heat from passing directly to the enclosure.

2 Assemble the two temperature-controlled zones. Secure the front dual-tube metal block and main-chamber heating assembly separately. Fit the heating elements closely against their contact surfaces and secure the temperature sensors. Keep the two zones separate, label their harnesses individually and leave the fan outlets clear.

3 Install the rotation mechanism. Connect the actuator to the three-sided working chamber, then check for collisions and binding throughout its permitted travel. Label the three viewing faces A, B and C, and label the tube positions on each face 1 to 5 so they can be matched to the acquisition records.

4 Install the optical components. Position the USB camera, lens and optical filter components in front of the viewing face, with blue LEDs for image acquisition. Check each face to confirm that all five tube positions are fully visible, in focus and unobstructed. Secure the bracket and leave enough slack in the moving harnesses.

5.2 Main control board fabrication and installation

Create the schematics and PCB layouts in JLCEDA, then export the Gerber and drill files to fabricate the core, control, mechanical and power boards. Before assembly, check the component packages, polarity and connector orientation. Complete the soldering checks on each board first.

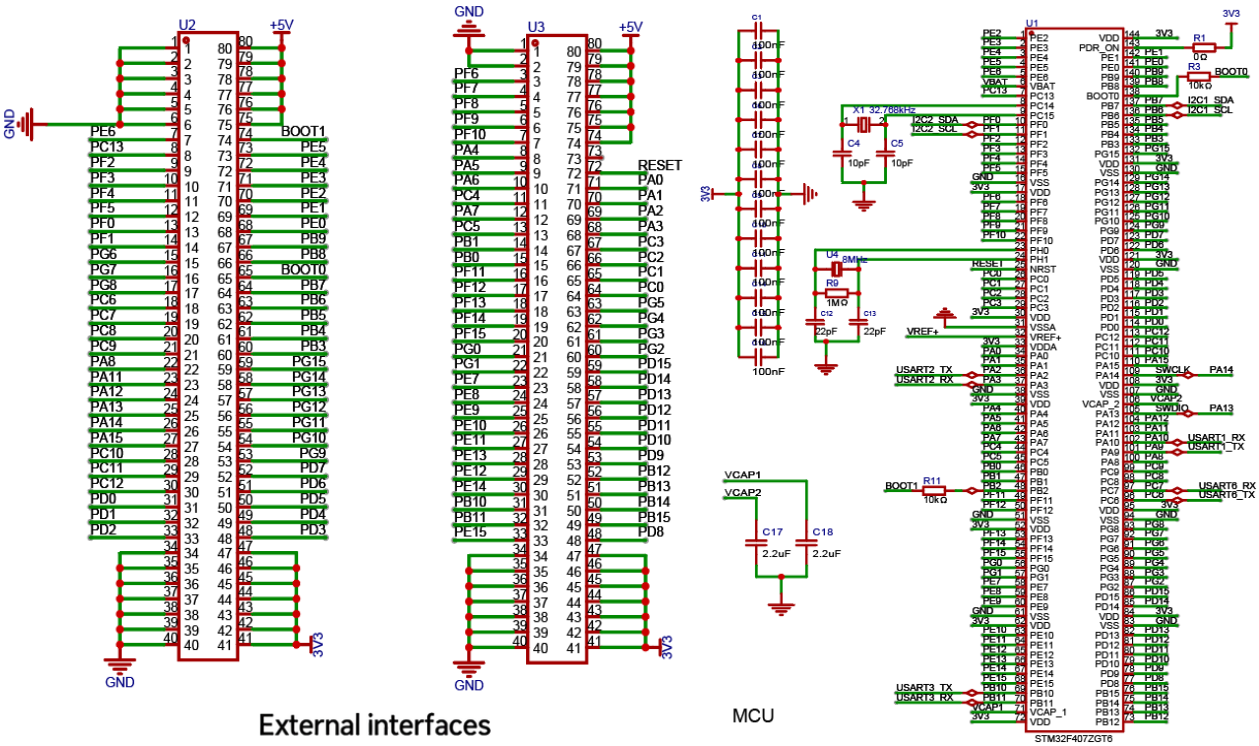


Figure 2 Inter-board connectors and STM32 core circuit

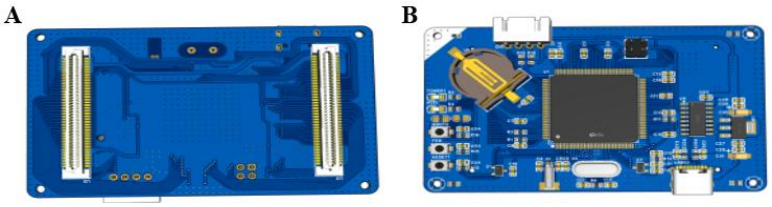


Figure 3 Connector side and component side of the core-board PCB

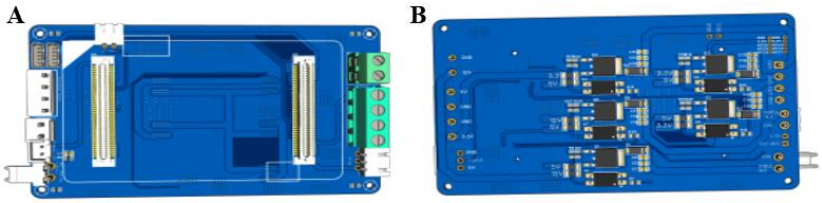


Figure 4 Interface and driver layout on the control-board PCB

The core board contains the STM32F407ZGT6 and its power, clock and reset circuits. It connects to the control board through inter-board connectors U2 and U3. Check pin 1 and connector orientation against Figure 2. Secure the board with insulating supports before mating the connectors to prevent misalignment.

The SWD interface CN1 has the following pinout: pin 1, 3V3; pin 2, SWDIO; pin 3, SWCLK; pin 4, GND. Connect a compatible programmer, confirm the target device and boot configuration, then flash the firmware. Verify temperature sensing and serial communication first.

5.3 Heating and lighting drivers

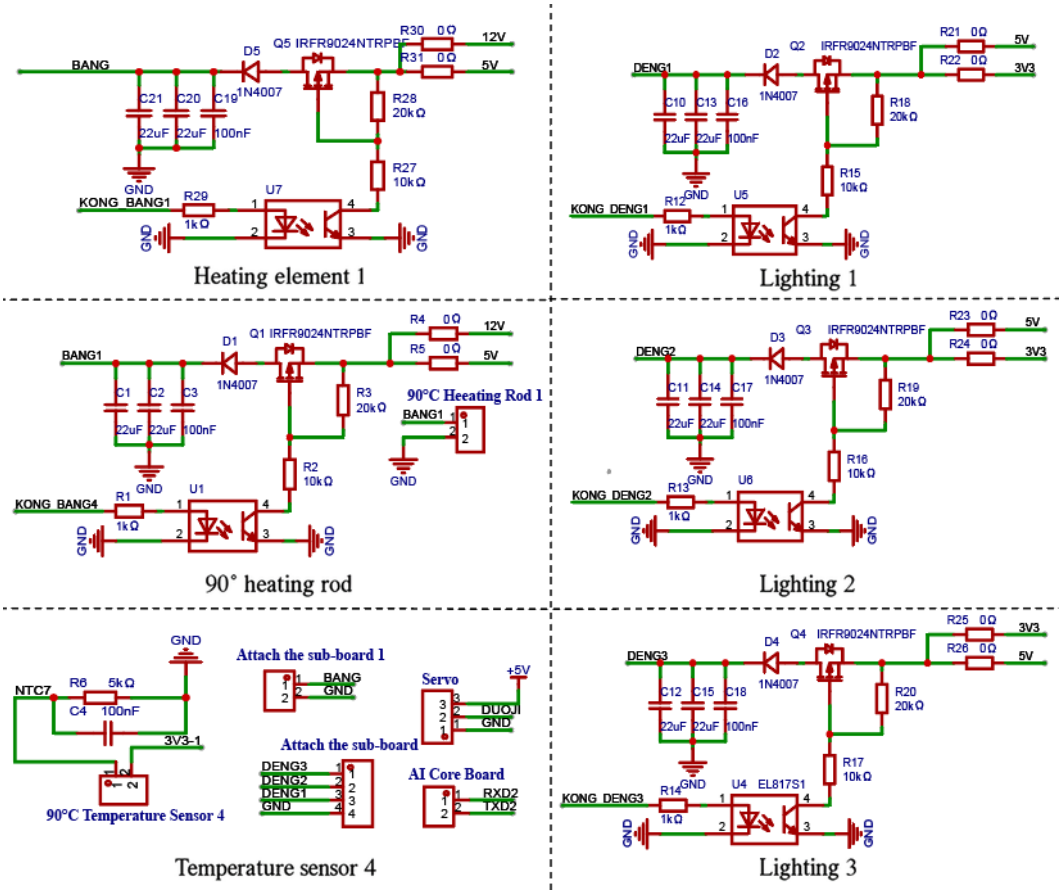


Figure 5 Heating circuits for the two temperature-controlled zones, three lighting channels and actuator interfaces

The control board sends signals from the core board to MOS driver circuits that separately regulate heating in the main chamber and front zone, and the three lighting channels. The driver supply is selected with 0 Ω resistors. Choose the appropriate branch for the actual load's rated voltage before soldering. Never fit both selection resistors in the same group.

Output net	Selection branch 1	Selection branch 2
Main chamber BANG	R30 → 12 V	R31 → 5 V
Front zone BANG1	R4 → 12 V	R5 → 5 V
Lighting DENG1	R21 → 5 V	R22 → 3V3
Lighting DENG2	R23 → 5 V	R24 → 3V3
Lighting DENG3	R25 → 3V3	R26 → 5 V

The resistor arrangement for the third lighting channel is reversed relative to the first two; identify the branches by their net names. Disconnect the loads and check the output voltage and polarity first. Then connect the heating elements, LED boards and actuators, confirming each controlled component in turn.

5.4 Mechanical board and peripheral connections

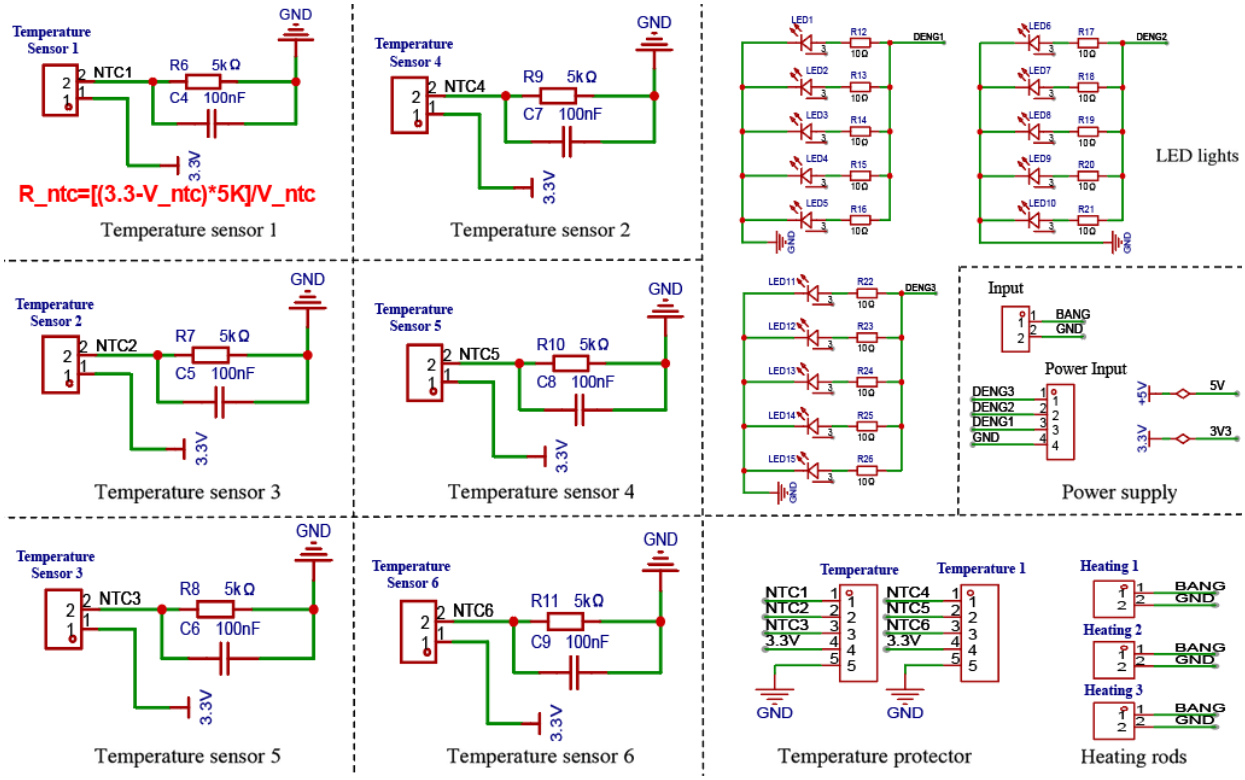


Figure 6 Six temperature-sensing channels, three lighting groups and main-chamber heating circuits on the mechanical board

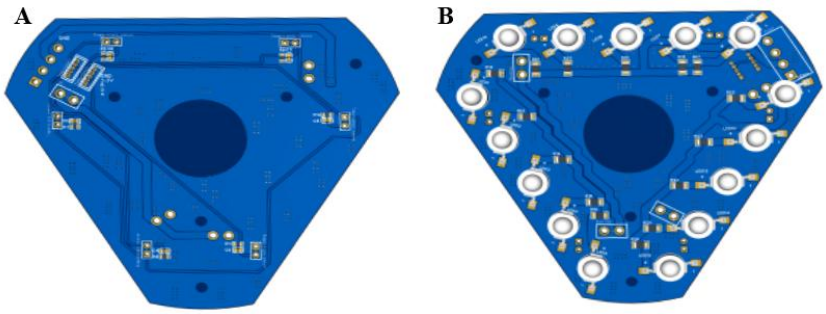


Figure 7 Routing side and component side of the mechanical-board PCB

The mechanical board has six NTC temperature-sensing branches and three groups of five LEDs. The sensor voltage-divider signals feed into the control board, and each lighting group corresponds to a viewing face. The three heater connectors share the BANG net and receive power from the same driver branch.

1 Secure the board and connect the wiring. Mount the mechanical board as shown in Figures 6 and 7. Label the sensors and harnesses NTC1 through NTC6, and connect the corresponding heating elements and lighting groups. Keep the board and wiring clear of the tube loading and unloading path and the camera field of view.

2 Check clearance for rotation. Route the harnesses along the bracket and secure them at their base, leaving enough length for rotation. Rotate through each face and confirm that the wires do not pull on the connectors, wind around the shaft or obstruct the tube positions.

5.5 Power components and supply checks

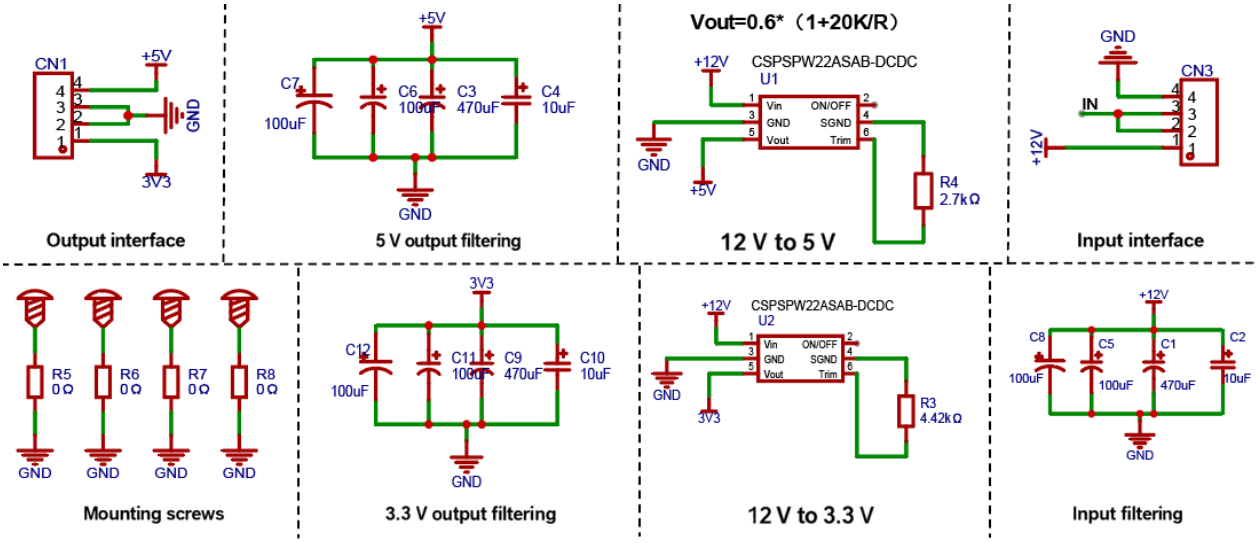


Figure 8 Power-board input, voltage conversion, filtering and output nets

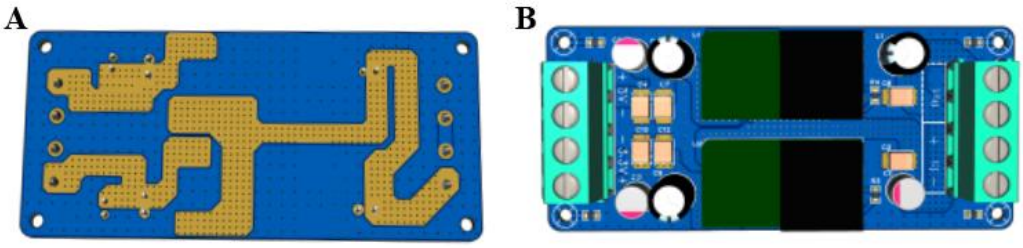


Figure 9 Power-board PCB routing and component placement

The battery, balancing and protection components, charging management components and power board work together to supply the computing, temperature-control and optical modules. Secure each component in its designated position. Keep the battery disconnected while checking input polarity, the protection path and the output nets.

The pin order of CN1 on the power board differs from that of the power connector on the control board. Wire them according to the table below. Before applying power, check for unintended shorts between each supply rail and ground. Measure the outputs without a load before connecting the functional boards.

Power-board CN1	Net	Control-board power connector
Pin 4	+5 V	Pin 1
Pin 2 or 3	GND	Pin 2 or 3
Pin 1	3V3	Pin 4

Power up in this order: logic boards, camera and sensors, lighting, actuators, then heating. Check the supply and operating status each time a load is added. If the voltage drops or the instrument restarts, disconnect that branch and investigate it first.

5.6 Inter-board wiring and control mapping

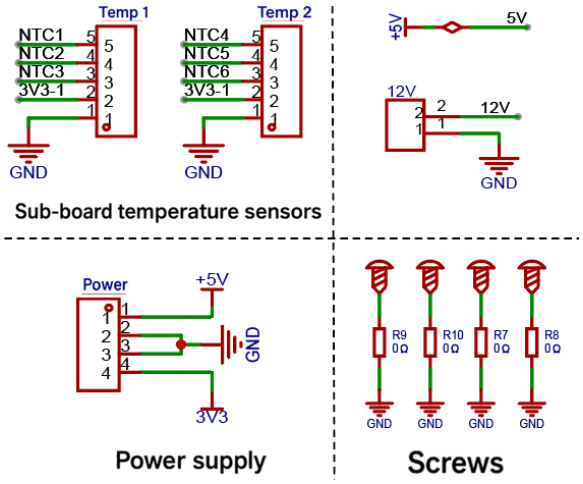


Figure 10 Control-board temperature inputs and power-connector pin definitions

The table below lists the connections by schematic pin number. Before crimping, check the viewing side of each connector, identify pin 1 and verify continuity. Wire by net name; the arrows do not indicate direct connections between identically numbered pins.

Circuit	Connections
Temperature-sensing group 1	Mechanical-board Temperature pins 1/2/3, carrying NTC1/2/3, → control-board Temperature 1 pins 5/4/3, respectively; mechanical-board pin 5 GND → control-board pin 1 GND.
Temperature-sensing group 2	Mechanical-board Temperature 1 pins 1/2/3, carrying NTC4/5/6, → control-board Temperature 2 pins 5/4/3, respectively; mechanical-board pin 5 GND → control-board pin 1 GND.
Temperature-sensing supply	Both groups on the mechanical board use pin 4 for 3.3V; both groups on the control board use pin 2 for 3V3-1. Verify the reference relationship between these two nets before wiring.
Main-chamber heating	Control-board Attach the sub-board 1 and mechanical-board Input: pin 1 BANG, pin 2 GND.
Three lighting groups	Control-board Attach the sub-board and mechanical-board LED Power Input: pin 1 DENG3, pin 2 DENG2, pin 3 DENG1, pin 4 GND.
Front zone and servo	Front-zone heating: pin 1 BANG1, pin 2 GND. Servo: pin 3 +5V, pin 2 DUOJI, pin 1 GND.
Computing-board serial port	AI Core Board interface: pin 1 RXD2, pin 2 TXD2. Connect these to the computing board's TX/RX, respectively, and check the signal levels and common ground.

Schematic pin mapping: main-chamber heating, PA0; front-zone heating, PB11; lighting channels 1 to 3, PB3, PE13 and PE14; servo, PB10; NTC1 through NTC6, PA1 through PA6; NTC7, PA7; RXD2, PD6; TXD2, PD5.

Supporting modules and interface identification

Use the following photographs to identify the supporting modules and their interfaces for the second-generation instrument. Before connecting them, match the board markings and component specifications to the corresponding software readout method.

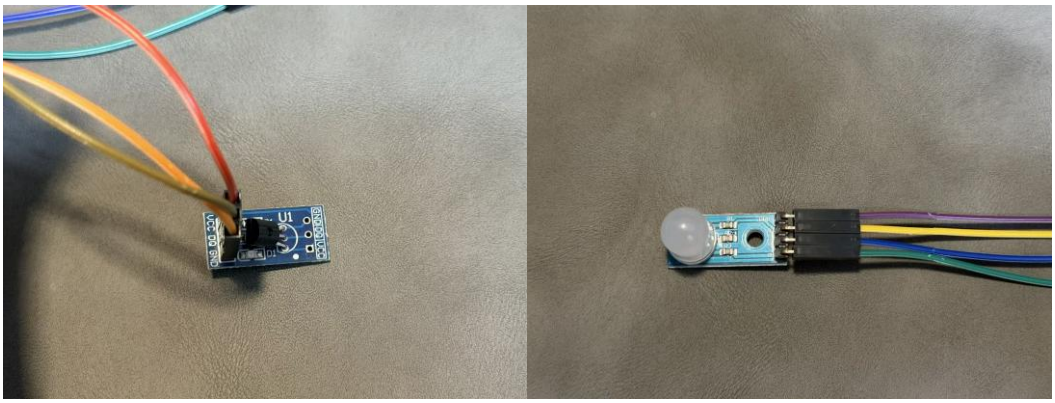


Figure 11 Photographs of the second-generation temperature-sensing and lighting modules. Left: temperature-sensing module with VCC, DQ and GND terminals. Right: RGB LED module with GND, R, G and B terminals.

DQ on the temperature-sensing module is a digital signal interface and requires the appropriate readout program. Do not apply the NTC voltage-divider wiring shown in Figure 6 directly to this module. Identify the RGB module's individual color channels, and check its intended use separately from the excitation-lighting circuits in Figure 5.

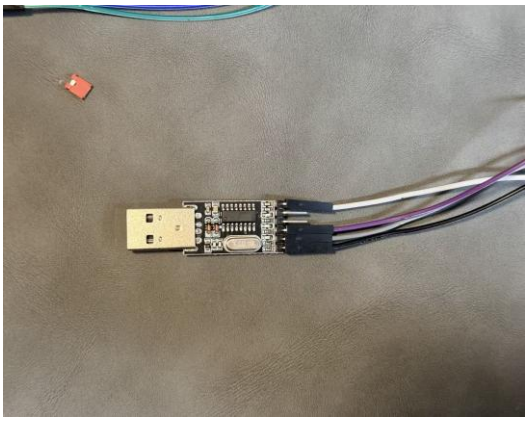
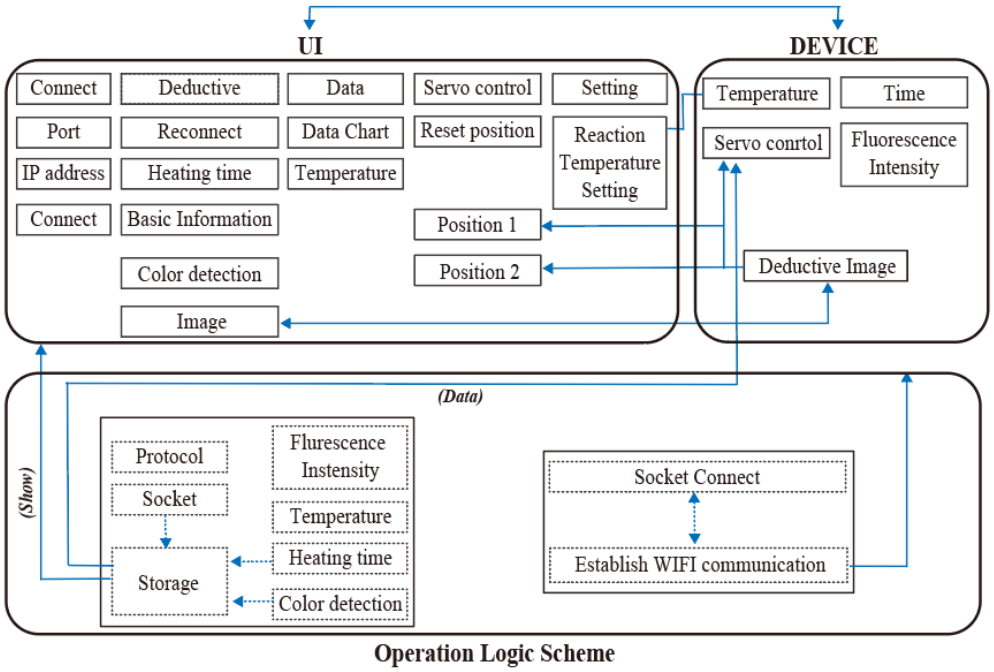


Figure 12 Photograph of the second-generation USB-to-serial module and Dupont jumper wires, showing the USB connector, pin header and detachable wiring harness.

For serial connections, check the module's signal levels and pin-header definitions. Cross-connect TX and RX to the target device and connect their grounds. Use the USB-to-serial module separately from the SWD programming interface in Section 5.2. Determine the specific configuration from the component and program versions.

5.7 Software setup and app connection



Operation Logic Scheme

Figure 13 Control, acquisition and data-storage workflow between the app and instrument

1 Configure the embedded controller. Map temperature sensing, heating in both zones, the three lighting channels, the servo and the serial port in the STM32 program. Confirm that the channels can be read and the actuator outputs can be controlled before connecting the program running on the RK3588.

2 Configure image processing. Connect the USB camera to the computing board and acquire raw images first. Then integrate YOLOv8n and OpenCV for localization and signal extraction. Deploy the model through the PT → ONNX → RKNN conversion path, keeping input preprocessing and output labels consistent.

3 Connect the mobile device. Connect the phone and instrument to the same local network. Enter Server IP and Port in the app, then select Connect to Server. Once new data arrive, check the temperature, time and current viewing face before sending an acquisition request.

4 Link positions to records. Associate faces A/B/C and tube positions 1 to 5 on each face with the image records. Acquire an image after each face has reached its position. Save the time, face identifier, raw image and analysis results, avoiding cached data from the previous face.

5 Check mobile-device compatibility. On each phone operating system intended for use, check connection, control, acquisition and saving. Test reconnection after a dropped connection and configuration restoration after a restart. Confirm that records continue to update.

5.8 Integrated system testing and maintenance

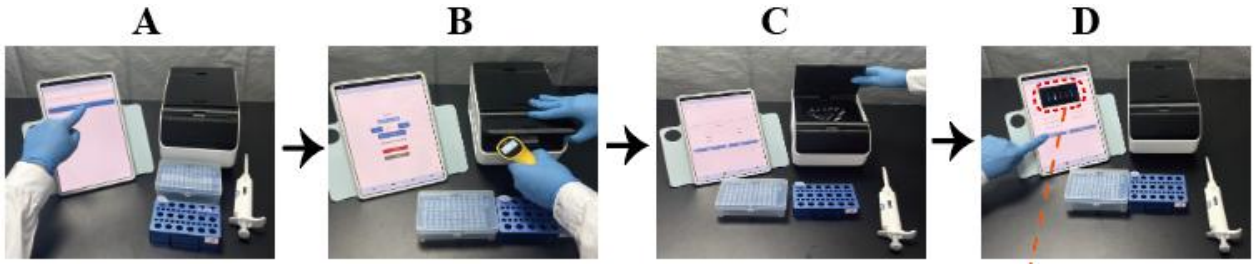


Figure 14 Photographs of integrated testing of byegerm2.0 with a mobile device. A: connecting the instrument and mobile device. B: operation at the front. C: operation at the top. D: viewing the results screen.

1 Verify the temperature channels. Start with readings at room temperature. Match each sensor to its software channel and compare each measurement point with a reference thermometer. Enable heating only after confirming the wiring, mounting positions and conversion calculations.

2 Test the two temperature-controlled zones. Check warm-up, steady operation and the response after stopping for the main chamber and front zone separately, then run both together. Confirm that setting changes affect the intended zone without causing power problems or interference between channels.

3 Check positioning repeatability. Switch to faces A, B and C in sequence, then return to the starting face. Compare focus and full visibility across the five tube positions on each face. After repeated switching, the physical position should match the recorded face identifier. Wait until movement has stopped and the chamber is stable before acquiring an image.

4 Complete integrated acquisition. Keep the reference object, exposure and lighting conditions fixed while running temperature control, rotation and imaging together. Check localization at each tube position and the file numbering. Save the temperatures, images and program configuration as the maintenance baseline for this instrument.

Symptom	Check first
Instrument will not start or restarts during operation	Input power, output voltages, common ground and newly added loads
Abnormal temperature or mismatched channels	NTC voltage dividers, reversed connector pin order, mounting points and software channels
Wrong heater or lighting group operates	BANG/BANG1 and DENG1 to DENG3 harnesses, and firmware mapping
Rotation binds or images shift	Mechanical interference, moving harnesses, actuator mounting and positioning parameters
Image does not match the face identifier	Positioning and acquisition sequence, timestamps, cache and record numbering
No new data in the app	Local network, address and port, instrument services and serial communication

Stop the instrument and disconnect power before maintenance. After replacing temperature-control, imaging or rotation components, recheck the affected modules. Then acquire images from all three faces and test the two temperature-controlled zones together.